RoboGambit 2025–26

Hardware Rulebook

A Physical Robotics Competition Focused on Perception, Manipulation, and Autonomous Decision Making

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1 Introduction

The RoboGambit Hardware Stage represents the physical execution phase of the RoboGambit competition. In this stage, teams deploy their perception, decision-making, and robotic manipulation pipelines on a real-world robotic system to participate in a chess-inspired strategic game.

The Hardware Stage builds upon the algorithms and pipelines developed during the Simulation Stage. Teams are expected to integrate perception, game reasoning, and robotic manipulation to operate reliably within a real-world environment.

The competition arena consists of a 6×6 game board flanked by two robotic arms used to manipulate cylindrical game pieces. Teams must use vision-based perception, autonomous decision-making, and precise manipulation to complete the tasks defined in this stage.

The Hardware Stage evaluates the ability of teams to translate simulation-developed algorithms into reliable real-world robotic execution.

2 Eligibility and Participation

2.1 Qualification

Participation in the Hardware Stage requires successful completion of the Software Stage submission requirements.

Only teams that have:

- Submitted the required software stage deliverables
- Achieved the minimum qualifying threshold

will be eligible to participate in the Hardware Stage.

2.2 Frozen Game Engine

The game engine submitted during the software stage will be considered **frozen**. Teams are not permitted to modify the internal decision-making logic, heuristics, or trained models.

Teams may modify only the following components to adapt their systems to hardware:

- perception pipelines
- input/output interfaces
- robot control pipelines

Any modification to the core game engine logic may result in disqualification.

2.3 Team Structure

Team composition rules remain identical to those defined in the official RoboGambit rulebook.

For the Hardware Stage, teams may include **one additional member** to act as the designated human chess player during matches.

3 Hardware Platform

3.1 Robotic System

Each competition arena includes the following hardware components:

- Two 4-DOF robotic arms
- Electromagnetic end effectors
- ESP32 microcontrollers for communication
- Control electronics for solenoid actuation

Teams are not permitted to modify the hardware setup.
All system behaviour must be implemented through software.

3.2 Arena Setup

The competition arena contains:

- a 6×6 grid game board
- a piece collection area
- fixed robot mounting locations

An overhead camera provides a global view of the arena.
All perception must be performed using the video feed from this camera.

3.3 Arena Layout and Dimensions

The competition arena consists of a square game board used for the RoboGambit mini-game along with the robotic manipulation setup. The following figure shows the layout of the competition arena. The final arena used during the competition will follow the same structural arrangement.

Board Dimensions

The game board used during the Hardware Stage has the following dimensions:

- Board size: **360 mm** $\times$ **360 mm**
- Grid: 6×6 **squares**

Each square therefore has an approximate size of 60 mm $\times$ 60 mm.

Board Coordinate System

The board squares follow a coordinate naming system similar to a standard chessboard.

- The **top-left square is labelled A1**.
- Square **A1 is a black square**.
- Columns progress from **1 to 6** from left to right.
- Rows progress from **A to F**.

This coordinate system is consistent with the board representation used during the Software Stage.

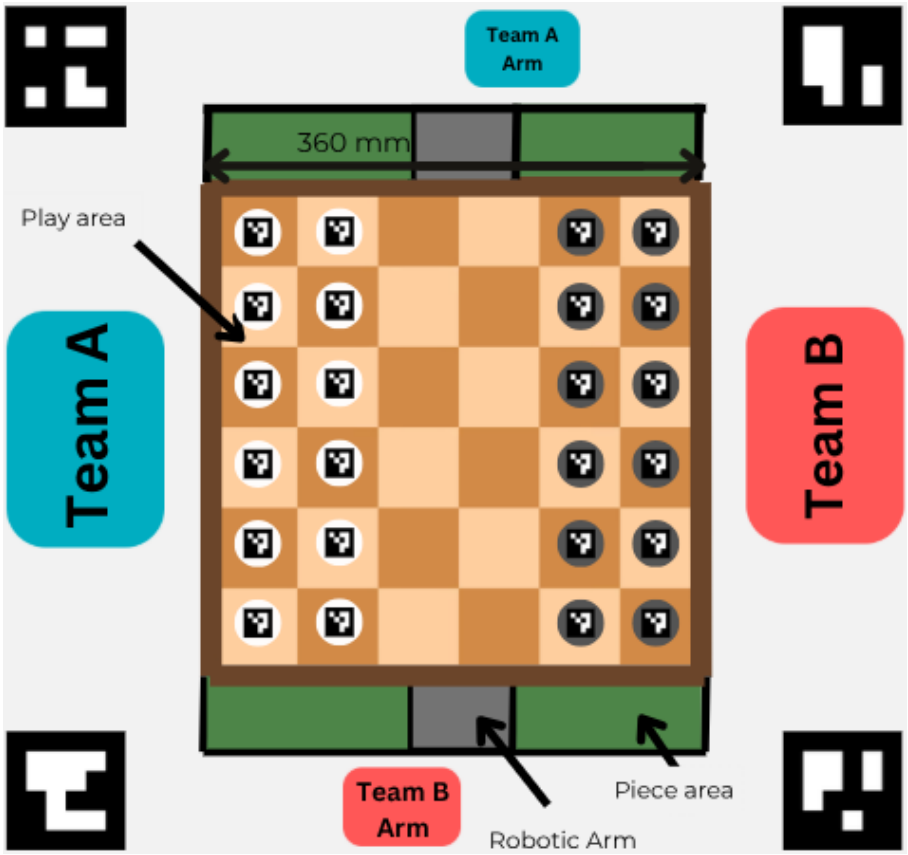


Figure 1: Updated RoboGambit arena layout showing the 6×6 board and camera placement.

Camera Placement

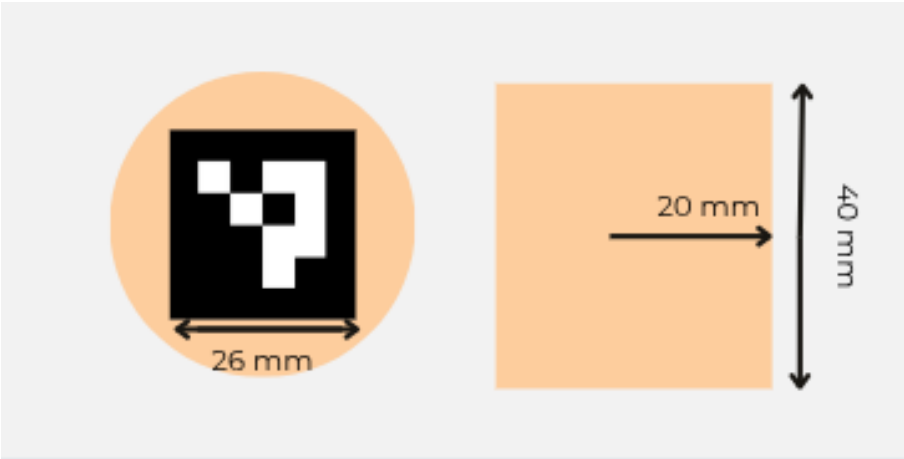
An overhead camera will be mounted directly above the arena to provide a global view of the game board.

- The camera will be positioned approximately above the **center of the arena**.
- The camera axis will face directly downward toward the board.
- Teams must use this camera feed for perception and board state estimation.

3.4 Game Pieces

The game pieces used in the competition have the following properties:

- Shape: Cylindrical
- Diameter: 40 mm
- Height: 40 mm
- ArUco marker mounted on top surface



These markers are used for vision-based detection and identification.

4 Hardware Access Slots

4.1 Testing Slots

Teams will be provided with dedicated hardware access slots for testing and debugging their systems.

During these slots, teams may:

- test perception pipelines
- calibrate robot control
- debug integration between software and hardware

The schedule for these slots will be released separately by the organising team.

4.2 Slot Rules

- Teams must operate only during their assigned slot.
- Each team must bring their own computation system.
- Hardware modifications are strictly prohibited.
- Organisers will supervise hardware operation.

Damage to the hardware setup may result in penalties or disqualification.

5 Hardware Tasks

The Hardware Stage consists of a single gameplay phase where teams participate in a chess-inspired game using a combination of human and autonomous robotic actions.

5.1 Collaborative Gameplay Phase

Matches are played in a **one-versus-one format**.

- The team assigned the **white pieces** will make the first move.
- The team assigned the **black pieces** will make the second move.

- Piece colour assignment may be determined by **seeding or random selection**.

For each team, turns alternate between the robotic system and the human player.

Autonomous Turn

The first move for each team will be executed by the **autonomous game engine**.

This turn is initiated by a designated team member referred to as the **Arm Player**. The Arm Player is only permitted to run a script named `main.py`.

The `main.py` script must contain the complete autonomous pipeline, including:

- perceiving the board state from the camera feed
- inferring the next move using the team's game engine
- executing the move using the robotic arm (involves controlling the electromagnet)

No additional manual intervention is permitted during this process.

Human Turn

Following the autonomous turn, the next move for the team is executed by the **Human Player**.

The Human Player physically performs a valid move on the board.

5.2 Player Interaction Restrictions

During a match, the **Arm Player** and the **Human Player** are not permitted to communicate or interact with each other.

Each player must perform their respective role independently during their turn.

Move Order

The sequence of moves during a match follows a fixed alternating pattern between autonomous and human turns for both teams.

Step	Team	Turn Type
1	White	Autonomous
2	Black	Autonomous
3	White	Human
4	Black	Human

This sequence repeats for the duration of the match.

5.3 Turn Completion

A move is considered complete once:

- the piece has been successfully placed on the intended square
- the robotic arm has released the piece (for autonomous turns)

After the move is completed, a referee will verify the board state and give the signal for the opposing team to begin their turn.

5.4 Captures, Promotions, and Move Logging

Special procedures apply for piece captures and pawn promotions during gameplay.

Piece Captures

If a piece is captured during a move executed by the robotic arm, the robot must remove the captured piece from the board and place it **outside the arena** in the **designated piece collection area closest to the robotic arm**.

The captured piece must not remain within the board region after the move is completed.

Pawn Promotion

In the event of a pawn promotion, the following procedure must be followed:

- The pawn being promoted must be removed from the board and placed **outside the arena** in the designated piece collection area.
- During a **human turn**, the Human Player will place the promoted piece on the promotion square.
- During an **autonomous turn**, the Arm Player must place the promoted piece as indicated by the team's game engine.

The chosen promotion piece must correspond to the move produced by the game engine and may be verified by the referees.

Move Logging

Teams are required to **log all moves performed during the match** in the same format prescribed during the Software Stage.

This includes:

- all standard moves
- capture moves
- promotion moves

Promotion moves must clearly indicate the **selected promotion piece**. The move logs may be inspected by the referees to verify correct execution and compliance with the game engine output.

6 Hardware and Software Resources Provided by Organisers

During the Hardware Stage, the organising team will provide the physical robotic infrastructure required to interact with the competition arena. Teams are expected to integrate their software pipelines with these systems during their assigned testing slots and during the competition matches.

6.1 Robotic Manipulation Hardware

Each practice setup will include the following robotic components:

- One 4-DOF robotic arm mounted on the side of the arena
- Electromagnetic end effector for piece manipulation, with an aruco tag pasted on top
- ESP32 microcontroller used for communication and coordination

- Relay module for controlling the electromagnetic gripper

Teams are not permitted to modify the provided hardware.

All control behaviour must be implemented through software interfaces provided by the organisers.

6.2 Perception Infrastructure

An overhead camera will be mounted above the arena to provide a global view of the board.

- Camera resolution: approximately 1920×1080
- Frame rate: approximately 30 FPS

The camera feed will be made available to teams for perception and state estimation.

Teams are expected to process this video stream using the perception pipeline developed during the software stage.

6.3 Resources Provided

To enable teams to integrate their systems with the competition hardware, the organisers will provide the following resources:

- Access to the overhead camera video feed for perception and board state estimation
- Documentation for the **Waveshare RoArm M2** robotic arm used in the competition setup
- Starter code intended to assist teams in interfacing with the hardware platform

The provided materials are meant only to assist teams in getting started with the hardware platform. Teams are responsible for implementing their own perception pipelines, decision-making integration, and robot control logic required to complete the competition tasks.

6.4 Use of Software Stage Pipelines

Teams are expected to reuse the core software components developed during the Software Stage.

In particular, the **game engine implemented in `game.py`** and submitted during the Software Stage must be reused for the Hardware Stage. Teams are not permitted to modify the internal decision-making logic of the game engine.

Teams may directly reuse the **perception node developed for ArUco detection and board state estimation** with the real camera feed provided during the Hardware Stage.

Additional scripts, nodes, or programs may be written to establish a pipeline between the perception system, the game engine, and the robotic arm used for manipulation.

Teams are responsible for integrating these components and adapting their software pipelines to operate reliably with real-world sensor inputs and the provided hardware platform.

7 Match Rules and Gameplay Regulations

7.1 Time Control

Each team receives a **total time of 10 minutes** for the entire match. In the upper stage matches (Upper Final, Lower Final and the Grand Final), the teams will get **15 minutes**.

- The clock runs only during the team's turn.

- The time is shared between the human player and the robotic system.

There is **no per-move time limit**.
If a team exhausts its allotted time, it immediately loses the match.

7.2 Draw and Tie-Break (Armageddon)

In the event that a match results in a **draw**, an **Armageddon tie-break** game will be played to determine the winner.

Bidding Process

Before the Armageddon game begins, both teams must secretly bid a portion of their total match time.

- The team that bids the **higher time** will be assigned the **black pieces**.
- The bid amount will be **subtracted** from that team's total available time.
- The opposing team (white) retains the **full allotted time**.

Winning Conditions

- The team playing as **white** must achieve a **win** to be declared the winner.
- The team playing as **black** is declared the winner in case of:
  - a **win**, or
  - a **draw**

This ensures a decisive outcome for every match.

7.3 Illegal Moves, Piece Disturbance & Incomplete Moves

Illegal moves are handled according to standard chess rules.
If an illegal move is performed:

- the board state will be restored
- the team must make a legal move

The team's clock continues running during correction.
If a robot unintentionally disturbs pieces other than the intended one:

- the referee restores the board state
- the move must be replayed

A move is considered complete only when:

- the piece is placed on the correct square
- the robot releases the piece

If a move fails, the team may retry while the clock continues running.
If a team fails to perform a valid move (including piece disturbance and illegal moves) more than 3 times, they will be considered as having lost the match.

7.4 System Failures

If a team's system stops responding:

- teams may restart their software
- the match clock continues running

Human intervention with the robot is not permitted.

8 Competition Format

The Hardware Stage will follow an **8-team double-elimination bracket format**.

All qualified teams will be seeded into the bracket based on their performance in the Software Stage.

Seeding Criteria

Seeding will be determined using the **ELO ratings** obtained from the evaluation of each team's autonomous game engine during the Software Stage.

- Higher ELO rating corresponds to a higher seed.
- Seeding ensures balanced matchups in the initial rounds.

Bracket Structure

- All teams begin in the **winners bracket**.
- Teams that lose a match move to the **losers bracket**.
- A team is eliminated only after losing **two matches**.
- Final standings are determined based on progression through the bracket.

9 Safety Guidelines

All teams must strictly follow safety guidelines.

- Teams must not physically interact with robots during autonomous execution.
- Unsafe behaviour may result in penalties.
- Organisers may halt execution if unsafe conditions are observed.
- Before and after every move, ensure that the robotic arm returns to its init position.

Figure 2: 8-team double-elimination bracket used for the Hardware Stage.

10 General Clauses

- All submissions and systems must represent original work by the team.
- The organising committee reserves the right to verify system behaviour.
- The organisers may modify rules or schedules if necessary.
- Decisions made by the organising team are final.
- For any ambiguities in the rulebook, kindly reach out to the organisers asap.
- In case of any discrepancy, the Judge/Jury rule shall be considered final.